

Object Oriented Programming

CSIT — 2nd Semester — 2082 — Answer Sheet

Subject Code: CSC166

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. A book shop maintains the inventory of books that are being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the sales person inputs the title and author to the system. If it is not available, "Not found" message is displayed. If it is available, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise "Required copies not in stock" is displayed. Design a system using a class called books with suitable member functions and constructors. Use new operator in constructors to allocate memory space required.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define polymorphism? List its advantages, write program to perform basic to user defined data type conversion.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Distinguish between public and private inheritance. Illustrate the chain of constructors and destructors in derived class with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. List and describe the characteristics of object oriented programming.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write any two types of storage class? Write a program to calculate Power (a, b), using default arguments, if second argument is omitted, then calculate square of a.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain copy constructor with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Is multiple inheritance possible? List some advantages and disadvantages of inheritance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Assume that there are three animals in which snake make sound as "hiss", dog make sound as "raf" and cat as "meow". Now implement this concept using pure virtual function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What are the uses of try and catch block? Write a program to throw an exception if the string given by user does not start with vowel.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How do you access file randomly? How do you test errors during file operations?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define stream. Suppose a file named STD.DAT contains a list of objects of class Student having data members name and class. Now write a program to extract the name of students reading in class 5 from STD.DAT.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Explain the use of 'this' pointer. Why do we need function template?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.